INCH-POUND
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10 July 2017
SUPERSEDING
MIL-PRF-32307
11 September 2008

PERFORMANCE SPECIFICATION

VALVE, QUARTER TURN, TRIPLE-OFFSET, TORQUE SEATED, SHIPBOARD USE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers triple-offset (see 6.3.4) torque seated (TOTS), quarter turn valves for fuel oil, aviation fuel, lubricating oil, seawater, auxiliary seawater, freshwater, chilled water, and potable water piping system services for shipboard use. The associated hydraulic or electric gear operators/actuators are not covered by this specification.

1.2 Classification. TOTS valves are of the following types, body styles, seat styles, sizes, and flange styles as specified (see 6.2).

1.2.1 Types. TOTS valves are classified into two types based on materials of manufacture (see table I).

a. Type I – Non-ferrous

b. Type II – Ferrous

1.2.2 Body styles. The following TOTS valve body styles are covered by this specification. There are two styles (wafer and lugged) and two thicknesses for each (short and long), totaling four style options. Short is equivalent to the face-to-face dimension of a MIL-V-22133 rubber-lined butterfly valve. Long is equivalent to the face-to-face dimension of a MIL-V-24624 high performance butterfly valve.

a. Short wafer

b. Long wafer

c. Short lugged

d. Long lugged

1.2.3 Seat styles. The following TOTS valve seat styles are covered by this specification. Seating may be integral to the valve body or of removable design.

a. Integral

b. Removable

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AMSC N/A FSC 4820

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1.2.4 Sizes. TOTS valve sizes conform to the following nominal pipe sizes (NPSs) in inches:

a. 2½*	d. 4	g. 8
b. 3	e. 5	h. 10
c. 3½*	f. 6	i. 12

* Less common valve sizes, use of which should be minimized for standardization and cost considerations.

1.2.5 Flange styles. TOTS valves are to be designed for installation between the following flange styles:

a. ASME B16.5

b. 803-1385892

1.3 Part or identifying number (PIN). PINs to be used for TOTS valves acquired to this specification are created as follows as specified (see 6.2):

M	**32307**	**-**	**#**	**X**	**X**	**#**	**X**
Prefix for Military Specification	Specification Number	-	Type (see 1.2.1 and code below)	Body Style (see 1.2.2 and code below)	Seat Style (see 1.2.3 and code below)	Size (see 1.2.4 and code below)	Flange Style (see 1.2.5 and code below)

PIN Code Explanation:

Type	**PIN Code**	**Body Style**	**PIN Code**	**Seat Style**	**PIN Code**	**NPS Size (Inches)**	**PIN Code**	**Flange Style**	**PIN Code**
I	1	Short wafer	A	Integral	I	2½	025*	ASME B16.5	A
II	2	Long wafer	B	Removable	R	3	030	803-1385892	B
		Short lugged	C			3½	035*		
		Long lugged	D			4	040		
						5	050		
						6	060		
						8	080		
						10	100		
						12	120		

* Less common valve sizes, use of which should be minimized for standardization and cost considerations.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections
QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-15024	-	Plates, Tags, and Bands for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plates, Identification
MIL-B-24480	-	Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service
MIL-V-24624	-	Valves, Butterfly, Wafer and Lug Style, Shipboard Service

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-798	-	Nondestructive Testing, Welding, Quality Control, Material Control and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385892	-	Flanges, Bronze, Special for Use with Butterfly Valves

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ISA 75.02.01 - Control Valve Capacity Test Procedures

NSF/ANSI 61 - Drinking Water System Components – Health Effects

(Copies of these documents are available online at http://webstore.ansi.org/.)

AMERICAN PETROLEUM INSTITUTE (API)

API-STD-598 - Valve Inspection and Testing

API-STD-607 - Fire Test for Quarter-turn Valves and Valves Equipped with Nonmetallic Seats

(Copies of these documents are available online at www.api.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.34 - Valves Flanged, Threaded and Welding End

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24 Metric/Inch Standard

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A182/A182M - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service

ASTM A193/A193M - Steel, Stainless and Alloy-Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A351/A351M - Standard Specification for Castings, Austenitic, for Pressure-Containing Parts

ASTM A479/A479M - Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels

ASTM B148 - Standard Specification for Aluminum-Bronze Sand Castings

ASTM B150/B150M - Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

ASTM B865 - Standard Specification for Precipitation Hardening Nickel-Copper-Aluminum Alloy (UNS N05500) Bar, Rod, Wire, Forgings, and Forging Stock

ASTM F104 - Standard Classification System for Nonmetallic Gasket Materials

ASTM F467 - Standard Specification for Nonferrous Nuts for General Use

ASTM F468 - Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use

ASTM F2168 - Standard Specification for Packing Material, Graphitic, Corrugated Ribbon or Textured Tape, and Die-Formed Ring

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ASTM F2191 - Standard Specification for Packing Material, Graphitic or Carbon Braided Yarn

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 3506-1 - Mechanical Properties of Corrosion-Resistant Stainless Steel Fasteners – Part 1: Bolts, Screws and Studs

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. TOTS valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.4).

3.2 Materials. TOTS valve assembly materials shall conform to type I or type II in accordance with table I.

TABLE I. TOTS valve assembly recommended materials. 1/

Description	Materials description			
	Type I (non-ferrous)		Type II (ferrous)	
	Specification	Material	Specification	Material
Body and seat	MIL-B-24480 and ASTM B148, C95800	Ni-Al-Brz	ASTM A351/A351M, Grade CF8M	Stainless steel
Integral seat	MIL-B-24480 and ASTM B148, C95800	Ni-Al-Brz	ASTM A351/A351M, Grade CF8M	Stainless steel
Removable seat 2/	ASTM B865, N05500, or MIL-B-24480 and ASTM B148, C95800	Ni-Cu-Al or Ni-Al-Brz	ASTM A240/A240M, Type 316 ASTM A479/A479M, Type 316	S31600
Seal ring 3/	QQ-N-286, N05500, and ASTM F104	Composite: Ni-Cu-Al and Flexible graphite 3/	ASTM A182/A182M, Type F316 ASTM A351/A351M, Grade CF8M ASTM F104	Composite: 316SS and Flexible graphite 3/
Disc	MIL-B-24480 and ASTM B148, C95800	Ni-Al-Brz	ASTM A182/A182M, Type F316 ASTM A351/A351M, Grade CF8M	Stainless steel
Stem and stem-to-disc fasteners	ASTM B865, N05500	Ni-Cu-Al 4/	ASTM A479/A479M, Type XM 19	Stainless steel
Wetted fasteners	ASTM F468 and ASTM F467, Alloy 400/500	Ni-Cu or Ni-Cu-Al 4/	ISO 3506-1, Grade A4, or ASTM A193/A193M, Grade B8M	Stainless steel

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TABLE I. TOTS valve assembly recommended materials – Continued. [1/]

Description	Materials description			
	Type I (non-ferrous)		Type II (ferrous)	
	Specification	Material	Specification	Material
Other wetted and moving parts	MIL-B-24480 and ASTM B148, C95800	Ni-Al-Brz	ASTM A351/A351M, Grade CF8M	Stainless steel
	ASTM B150/B150M, C63000	Ni-Al-Brz	ASTM A182/A182M, Type F316 ASTM A351/A351M, Grade CF8M	316SS
	ASTM B865, N05500	Ni-Cu-Al [4/]	ASTM A182/A182M, Type F316 ASTM A351/A351M, Grade CF8M	Stainless steel
Stem seals	ASTM F2168 and ASTM F2191	Braided graphite and Graphite	ASTM F2168 ASTM F2191	Braided graphite and Graphite
Brackets and adapters	MIL-B-24480 and ASTM B148, C95800	Ni-Al-Brz	ASTM A182/A182M, Type F316 ASTM A351/A351M, Grade CF8M	Stainless steel
	QQ-N-281	Nickel-Copper	---	---

NOTES:

1/ Equivalent materials may be substituted as approved by the NAVSEA Technical Warrant Holder.

2/ Seawater service requires Ni-Cu-Al material.

3/ Other equivalent seal ring materials or designs may be as approved by the NAVSEA Technical Warrant Holder.

4/ Ni-Cu in accordance with QQ-N-281 or ASTM F468 and ASTM F467 may be substituted where Ni-Cu-Al tensile strength is not required.

3.2.1 Material assessment. All materials that may come in contact with potable water for human consumption shall be in accordance with NSF/ANSI 61.

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.3 Design and construction. TOTS valves shall be of the types, body styles, seat styles, sizes, and flange styles specified (see 1.2 and 6.2). The stem, disc, and seat shall have a double-offset (see 6.3.2) design. The third offset shall be achieved by inclining the seat and seal cone centerline with respect to the pipe and valve centerline in order to eliminate rubbing during 90 degrees of operation.

3.3.1 Valve body. The dimensions of type I and type II valve bodies shall be in accordance with ASME B16.34 for class 150 valves. The minimum wall thicknesses for type I and type II valves shall be calculated in accordance with the "Basis Equations for Minimum Wall Thickness" table of ASME B16.34 for class 150 valves. Note that the design criteria specified in ASME B16.34 applies only to ferrous materials and shall be modified to consider the stress limits for non-ferrous materials prior to use in the design of type I (non-ferrous) valves. Valves shall be designed for installation between flanges in accordance with 803-1385892 for non-ferrous (type I) valves and ASME B16.5, class 150 for ferrous (type II) valves. Wafer and lugged valves shall be manufactured with face-to-face dimensions in accordance with table II.

TABLE II. Valve face-to-face dimensions.

NPS (inches)	High performance butterfly equivalent MIL-V-24624 replacements (±0.0625 inch)	Rubber-lined butterfly equivalent MIL-V-22133 replacements (±0.0625 inch)
2½	1.875	1.625
3	1.875	1.625
3½	2.0	1.75
4	2.125	1.875
5	2.25	2.0
6	2.25	2.0
8	2.5	2.375
10	2.8125	2.563
12	3.1875	2.875

3.3.2 Actuator mounting bracket. The valve body shall have an integral provision for mounting the manual operator or motor actuator. An adaptive mounting bracket may be utilized to attach the actuator, provided it is approved by NAVSEA and the entire assembly, valve, mounting bracket, and actuator are successfully shock tested and vibration tested as specified (see 3.5.4 and 3.5.5). Requirements for mounting actuators, including bolting size and pattern, shall be as specified (see 6.2).

3.3.3 Valve seat. The valve seat shall be integral to, or removable from, the valve body.

3.3.4 Disc. The seal ring assembly on the disc shall be replaceable without removing the stem and disc assembly.

3.3.5 Stem. The stem shall be of one-piece construction. The stem shall be designed so that the failure of the stem-to-disc connection or internal failure of the stem shall not permit any portion of the stem to be ejected from the valve as a result of internal pressure (see 6.2 and 6.8). The stem seals shall be capable of being tightened while the valve is under pressure. The stem seals shall be made of graphite and braided graphite or graphite rings (see table I), or other sealing arrangement proven to be equivalent or better.

3.3.6 Position indication. The indication of the position of the disc shall be integral with the valve stem. The design shall ensure that the valve (and operator) cannot be assembled to falsely indicate the valve's open and closed position.

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3.3.7 Actuators. Enclosed worm gear-type manual actuators shall be available for all size valves. The maximum allowable tangential force, based on hand-wheel size, is provided in table III. This force applied to the manual actuator shall meet or exceed the maximum torque required to close and open the valve (see 6.2).

TABLE III. Maximum tangential hand-wheel force.

Hand-wheel diameter (inches)	Total tangential force (rim of hand-wheel) (lbs)
2.5	30
3	30
3.5	30
4	30
5	50
6	72
8	102
10	120
12	135

3.4 Physical requirements.

3.4.1 Pressure-temperature ratings. The pressure-temperature rating for TOTS type I valves shall be calculated in accordance with the ASME B16.34 non-mandatory Appendix B method used for establishing pressure–temperature ratings, method for class 150 – All Materials. The pressure-temperature rating for TOTS type II valves shall be in accordance with the “Ratings for Group 2.2 Materials” table in ASME B16.34 for class 150, standard class valves. Both type I and type II TOTS valves shall be rated at 100 °F.

3.4.2 Seat tightness. TOTS valves shall meet the resilient seat leakage criteria in accordance with API-STD-598 (see 4.5.9).

3.4.3 TOTS valve body marking. The following markings shall be permanently marked on the valve body:

a. Manufacturer name
b. Model number
c. Valve size in NPS
d. Valve pressure class in pounds per square inch gauge (psig) (optional)
e. Preferred flow indication (stem side)

3.4.4 Identification plate. TOTS valves shall be provided with a permanently attached identification plate in accordance with MIL-DTL-15024 and MIL-P-15024/5. The identification plate shall contain the following information:

a. Manufacturer name
b. Size of valve in NPS, pressure class in psig, and pressure rating in psig
c. Body material
d. Valve trim material: stem, disc, and seat
e. Contractor and Government Entity (CAGE) number
f. PIN (see 1.3)

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3.5 Performance requirements.

3.5.1 Flow capacity. The minimum flow capacity shall be in accordance with table IV.

TABLE IV. Minimum flow coefficient (Cv).

NPS (inches)	Rated Cv [1/]
2½	43
3	85
3½	114
4	180
5	371
6	732
8	1,472
10	2,480
12	3,790

NOTE:
[1/] Disc side direction Cv.

3.5.2 Valve operation. TOTS valves shall be capable of fully opening from the fully closed position in 35 seconds or less. TOTS valves shall also be capable of fully closing from the fully open position in 35 seconds or less. Regardless of actuation method, TOTS valves shall operate under the pressure conditions and operating torque values listed in table V.

TABLE V. Maximum operating torque values.

NPS (inches)	Running (in-lb)	Pressure direction [1/]	Max allowable to close (in-lb at 275 psig ΔP)	Max allowable to open (in-lb at 275 psig ΔP)
2½	320	SS	677	1,120
		DS	677	1,120
3	363	SS	1,049	1,720
		DS	1,049	1,720
3½	400	SS	1,421	2,320
		DS	1,421	2,320
4	451	SS	1,793	2,940
		DS	1,793	2,940
5	584	SS	3,134	6,162
		DS	3,134	6,162
6	451	SS	3,492	6,880
		DS	3,492	6,880
8	885	SS	4,833	9,534
		DS	4,833	9,534

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TABLE V. Maximum operating torque values – Continued.

NPS (inches)	Running (in-lb)	Pressure direction [1/]	Max allowable to close (in-lb at 275 psig ΔP)	Max allowable to open (in-lb at 275 psig ΔP)
10	1,354	SS	8,961	17,614
		DS	8,961	17,614
12	1,850	SS	13,023	25,626
		DS	13,023	25,626

NOTE:
[1/] SS: Stem Side Flow Direction (see 6.3.3);
DS: Disc Side Flow Direction (see 6.3.1).

3.5.3 Fire resistance. TOTS valves, with actuators installed, shall be fire resistant and shall meet the allowable through valve and external leakage rate acceptance criteria when tested in accordance with API-STD-607 (see 4.5.3).

3.5.4 Shock resistance. Valves and their actuators shall meet the grade A shock acceptance requirements of MIL-DTL-901 when tested as specified in 4.5.4.

3.5.5 Vibration resistance. Valves and their actuators shall meet the type I vibration requirements of MIL-STD-167-1 when tested as specified in 4.5.5.

3.5.6 Life cycle and throttling. The valves shall be designed for a life of 20,000 cycles and shall be capable of throttling in the 15 to 17 percent open position for a minimum of 12 hours (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the examination of 4.4 and the tests of 4.5.2 through 4.5.9 (see table VI) of the 4-inch, 8-inch, and 12-inch TOTS valves (see 4.5.3 regarding selection of fire test valve sizes) to qualify the TOTS valve sizes listed in 1.2.4. Type I, body style A or B, seat style R (if applicable) shall be used for qualification of non-ferrous valves. Type II, body style A or B, seat style R (if applicable) shall be used for qualification of ferrous valves. Successful completion of the tests will allow qualification approval to be extended to the smaller intermediate valve sizes (see 4.5.1), body style C and D, and seat style I.

4.2.1 Extension of qualification with automated actuators. Tests conducted on automated valves, consisting of a valve and actuator (electric, hydraulic, pneumatic) combination, shall be extended to qualify the same actuator when installed on larger size valves. Successful tests for any body style (short or long) wafer valve shall be extended to the other body style wafer and both lugged body styles of the same type.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.7, 4.5.8, and 4.5.9 (see table VI).

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TABLE VI. Examinations and inspections.

Test	Requirement	Verification	Qualification	Conformance
Examination	3.2 through 3.3.7, 3.4.3, and 3.4.4	4.4	X	X
Life cycle and throttling tests	3.5.2 and 3.5.6	4.5.2	X	
Fire test	3.5.3	4.5.3	X	
Shock test	3.5.4	4.5.4	X	
Vibration test	3.5.5	4.5.5	X	
Flow capacity test	3.5.1	4.5.6	X	
Functional check	3.5.2	4.5.7	X	X
Hydrostatic shell test	3.4.1	4.5.8	X	X
Seat tightness test	3.4.2	4.5.9	X	X

4.4 Examination. Each TOTS valve shall be examined for compliance with the requirements specified in table VI. Any redesign or modification of the manufacturer's standard product not qualified in accordance with 4.2 shall constitute rejection of the valves. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Methods of inspection.

4.5.1 Test valve sizes. TOTS valves of each of the 4-inch, 8-inch, and 12-inch sizes shall be subjected to the tests specified in 4.5.2 through 4.5.9. The 4-inch, 8-inch, and 12-inch sizes shall be used in lieu of the valve sizes specified in MIL-STD-798. Successful testing of the 4-inch size approves the 2½-inch, 3-inch, 3½-inch, and 4-inch sizes only. Successful testing of the 8-inch size approves the 5-inch, 6-inch, and 8-inch sizes only. Successful testing of the 12-inch size approves the 10-inch and 12-inch sizes only. Qualification requirements for any other sizes require prior NAVSEA approval. For fire test valve sizes, see 4.5.3.

4.5.2 Life cycle and throttling tests. The following life cycle and throttling tests shall be conducted to determine conformance with the requirements of 3.5.2 and 3.5.6. The sequence of these tests shall be determined by the manufacturer.

4.5.2.1 Life cycle test. The life cycle test shall be conducted using a test fixture that shall allow either side of the valve to be pressurized at the 100 °F rated pressure with clean, ambient temperature tap water when the valve is in the closed position. Once a side is pressurized, the valve shall be cycled to simulate operation in a shipboard system. A cycle consists of fully opening the valve from the fully closed position and returning it to the fully closed position. The valve shall undergo 20,000 cycles. Every 2,000 cycles, the valve shall be tested for leakage in both the stem side (SS) and disc side (DS) directions using clean, ambient temperature tap water at the 100 °F rated pressure. The test pressure shall be held for a minimum of 1 minute. The number of cycles and the associated leakage rate shall be recorded. Every 2,000 cycles, the flow direction shall be reversed to verify the bidirectional capabilities of the valve. The operating time shall not exceed 35 seconds. The torque to fully open the valve from the fully closed position and to fully close the valve from the fully open position shall not exceed the requirements of table V. The disc seal ring, the seat (if removable type is used), and the stem seal shall not be replaced during the test. During the cycle test, the stem seal may be adjusted. The water leakage rate shall not exceed the maximum seat leakage rate of 10 cubic centimeters per inch of NPS per hour. Galling or seizing of the moving parts shall constitute failure of the test.

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4.5.2.2 Throttling test. Each test valve shall be set in the 15- to -17 percent open position and clean tap water shall be passed through the valve for a minimum of 12 hours. The 4-inch, 8-inch, and 12-inch valve sizes shall be throttled through a 100-lb/in^2 differential. A new seat (if removable type is used) and new disc seal ring may be used for this test. Upon completion of this test, the valve shall be tested for leakage in both the SS and DS directions using clean, ambient temperature tap water at the 100 °F rated pressure. The test pressure shall be held for a minimum of 1 minute. The water leakage rate shall not exceed the maximum seat leakage rate of 10 cubic centimeters per inch of NPS per hour.

4.5.3 Fire test.

4.5.3.1 Procedure. TOTS valves shall be fire tested in accordance with API-STD-607. Test sizes shall be selected from the "Other Valves Qualified by NPS" table of API-STD-607. When selecting test sizes, consideration should be given to selecting the least number of valve sizes for fire testing to encompass a range of sizes of the same basic design that require qualification. During the test, type I valves shall be mated to flanges in accordance with 803-1385892. Type II valves shall be mated to ASME B16.5, class 150 raised face flanges. Leakage shall be measured by monitoring the source tank water levels during the test.

4.5.3.2 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is approved by the NAVSEA Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) or International Code Council's International Accreditation Services (IAS). The scope of accreditation shall include specific flammability and fire tests required for qualification. All other fire test provisions shall be as specified (see 6.2 and 6.7).

4.5.4 Shock resistance. Valves, with actuators installed (gear box, hydraulic, or electric), shall be tested in accordance with the grade A, class I, type A shock test requirements of MIL-DTL-901. For purposes of the shock test, the following MIL-DTL-901 classifications shall be applied to the TOTS valve:

a. Mounting location – deck or shell mounted as specified (see 6.2).

b. Mounting plane – in-line pipe.

c. Mounting orientation – unrestricted.

d. Test category – dependent on weight limitations for deck mounted valves; shell-mounted valves shall be heavyweight shock tested.

During the shock test, the valve shall be pressurized at the 100 °F rated pressure with ambient temperature water. The valve shall be tested both open with pressure throughout and closed with pressure on the disc side only.

4.5.4.1 Initial post-shock test. Initial post-shock test shall consist of an external visual and dimensional check without disassembling the valve, a hydrostatic shell test in accordance with 4.5.8, and a seat tightness test in accordance with 4.5.9.

4.5.4.2 Final post-shock inspection. Tested valves shall be disassembled and inspected for breakage, cracking, deformation, and misalignment. Highly stressed areas and seating areas shall be checked for cracks using a dye penetrant test in accordance with MIL-STD-798. A dimensional check of the valve and valve internals shall be performed to determine valve damage. The valve and valve actuator shall meet the requirements of 3.5.4.

4.5.5 Vibration test. Valves, with actuators installed, shall be tested in accordance with the type I requirements of MIL-STD-167-1. Testing shall be conducted while the valve is pressurized at 100 °F rated pressure with ambient temperature water, once in the closed position, then in the open position. There shall be no evidence of valve structural damage or performance degradation.

4.5.6 Flow capacity test. The flow capacity test shall be conducted in accordance with ANSI/ISA 75.02.01 to determine conformance with table IV. The test shall be conducted with the valve oriented in the non-preferred direction with the flow entering the disc side of the valve. Flow capacities of the intermediate valve sizes may be extrapolated, provided the flow path geometry is a scaled version of the test valve.

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4.5.7 Functional check. TOTS valves, with actuators installed, shall be cycled (fully open - fully close - fully open) a minimum of three times while pressurized at the 100 °F rated pressure with ambient temperature water. The operating time and operating torque shall be measured to determine compliance with the requirements of 3.5.2.

4.5.8 Hydrostatic shell test. Each valve shall be subjected to a body strength and mechanical assembly test at 150 percent of the 100 °F rated pressure. The test shall be performed with water at a temperature not less than 60 °F and the valve shall be in the open position. The test duration shall be a minimum of 3 minutes. Any external leakage shall constitute failure of this test and shall be cause for rejection.

4.5.9 Seat tightness test. Following the hydrostatic shell test, the valve shall be closed and the seat shall be tested for hydraulic and pneumatic tightness. The test duration shall be a minimum of 1 minute. The leakage rate shall not exceed the maximum leakage rate specified in API-STD-598 for resilient-seated valves when subjected to:

a. 110 percent of the 100 °F rated pressure, with ambient temperature water in the stem side flow direction.

b. 90 psig pressure with air in the stem side flow direction.

c. 110 percent of the 100 °F rated pressure, with ambient temperature water in the disc side flow direction.

d. 90 psig pressure with air in the disc side flow direction.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. TOTS valves are intended for flow control in fuel oil, aviation fuel, lubricating oil, firemain, seawater, and freshwater systems installed on Navy surface ships and submarines. TOTS valves are not approved for services requiring Level I and SUBSAFE levels of essentiality. Existing stock of TOTS valves manufactured to MIL-PRF-32307 dated 11 September 2008 are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required.

c. Type, body style, seat style, size, and flange style (see 1.2 and 3.3).

d. PINs required (see 1.3).

e. Applicable National Stock Number (NSN) if available.

f. Actuator to be supplied with the valve body: either (1) manual operator or (2) motor actuator, bolting size, and pattern required (see 3.3.7).

g. Requirement for evidence that the stem will not be ejected in any circumstance (see 3.3.5 and 6.8).

h. Additional fire testing provisions (see 4.5.3.2 and 6.7).

i. Shock test mounting location (deck or shell mounted) (see 4.5.4).

j. Packaging requirements (see 5.1).

k. Shelf-life requirements (see 6.5).

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6.3 Definitions.

6.3.1 Disc side (DS) flow direction. Flow acting in a direction on a closed valve that would unseat the valve disc from the valve body (also known as the “non-preferred flow direction” or “reverse flow direction” through the TOTS valve).

6.3.2 Double-offset. Characterized by offset of the valve stem behind the centerline of the valve sealing surface and an offset of the stem to the side of the centerline of the pipe.

6.3.3 Stem side (SS) flow direction. Flow acting in a direction on a closed valve that would seat the valve disc on the valve body (also known as the “preferred flow direction” or “normal flow direction” through the TOTS valve).

6.3.4 Triple-offset. Includes the double-offset design and incorporates a third offset, which acts to create a camming motion and effectively eliminates seat-to-seal rubbing throughout the valve’s 90-degree range of motion.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32307 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, Shelf-life Management Manual. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POCs). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil.

6.6 Valve seats. Material upgrade for valve seating should be used for severe service such as firemain and seawater service.

6.7 Additional fire testing provisions. NAVSEA reserves the right to witness the tests, and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements of the qualification tests. NAVSEA will provide a letter that indicates the approved laboratories which are accredited to ISO/IEC 17025 and which details how to obtain and maintain accreditation (see 6.2).

6.8 Stem integrity. Evidence for proof that the stem will not be ejected in any circumstance should consist of a detail in the control drawing showing the design consideration sufficiently detailed to document the design consideration for stem containment and an accompanying drawing note. Format in order of preference is: PDF, AutoCAD, Microsoft Word (or equal), hard copy (8½ inches by 11 inches), and drawings of larger format, if necessary. All submissions should be emailed to CommandStandards@navy.mil (see 3.3.5 and 6.2).

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6.9 Subject term (key word) listing.

Actuator

Fire resistant

Lugged

Manually operated

Shutoff, bidirectional

Wafer

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH

Preparing activity:
Navy – SH
(Project 4820-2017-002)

Review activities:
Navy – AS, SA
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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